LLMs

Language Models

Course Contents

Deep Learning
Transformer
LLMs
Context Engineering
Finetuning
Agents
The Bigger Picture

This Part: LLMs

- generative AI
- autoregressive text generation
- scaling laws
- LLM zoo

Generative vs Predictive/Discriminative Models

discriminative models:

predict conditional probability $P(Y|\boldsymbol{X})$

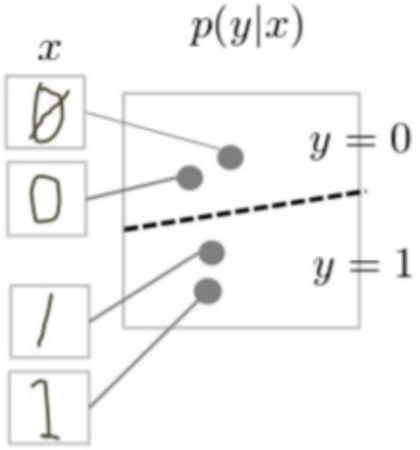
generative models:

predict joint probability $P(Y, \boldsymbol{X})$

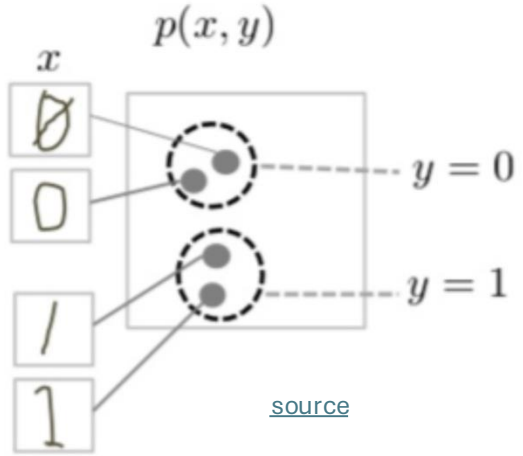
(or just $P(\boldsymbol{X})$ → unsupervised learning)

→ allow to generate new data samples

discriminative model



generative model



source

task of generative models more difficult: need to model full data distribution rather than merely find patterns in inputs to distinguish outputs

Generative models can be used for predictive tasks (Bayes theorem).
But predictive models are usually better at it.

Autoregressive Text Generation

LLMs (autoregressive transformers) generate one output token at a time (according to predicted probabilities)

then add it to the context for the prediction of the next token

→ resulting in a full text, sampled from predicted probabilities

How are LLMs Generative Models then?

at each step, take the sequence of previous tokens $x_{<t}$ (context) and predict a conditional probability distribution over the next token x_t:

$$P(x_t|x_{<t})$$

joint probability distribution over text sequences via chain rule of probability:

$$P(x_1, x_2, \cdots, x_T) = \prod_{t=1}^{T} P(x_t|x_{<t})$$

→ no direct prediction of joint probability in one shot, but implicit calculation through product of conditionals

Prompting

feed information into decoder-only transformer via input prompt

→ attention to context

using all the knowledge acquired during internet-scale pretraining and stored in weights

kind of programmable neural network

→ a new paradigm: **one model for many tasks**

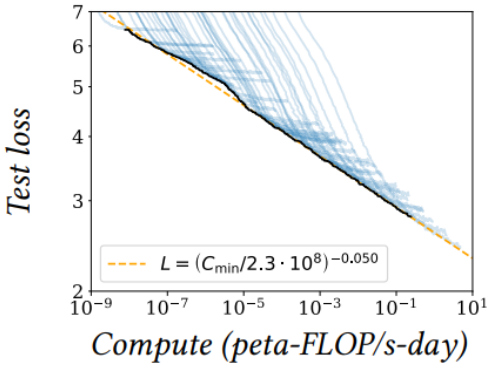
Size Matters: **LARGE** Language Models

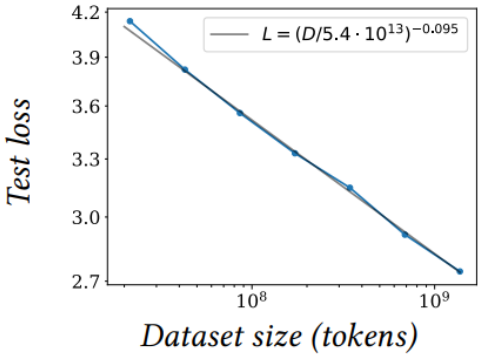
scaling laws, Chinchilla: coupled performance power laws with model size, amount of training data, and compute

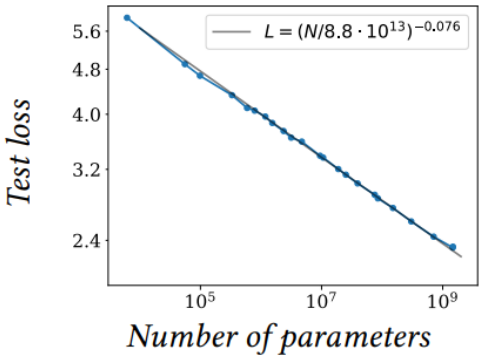
→ era of large-scale models

emergent abilities of LLMs:

- multi-task learning: perform new tasks at test time without task-specific training (via prompting)
- reasoning capabilities (improved by instruction tuning with multi-step reasoning examples)







source

Some LLM Numbers

example Kimi K2 (Mixture-of-Experts model):

as usual, comes in base and instruct variants

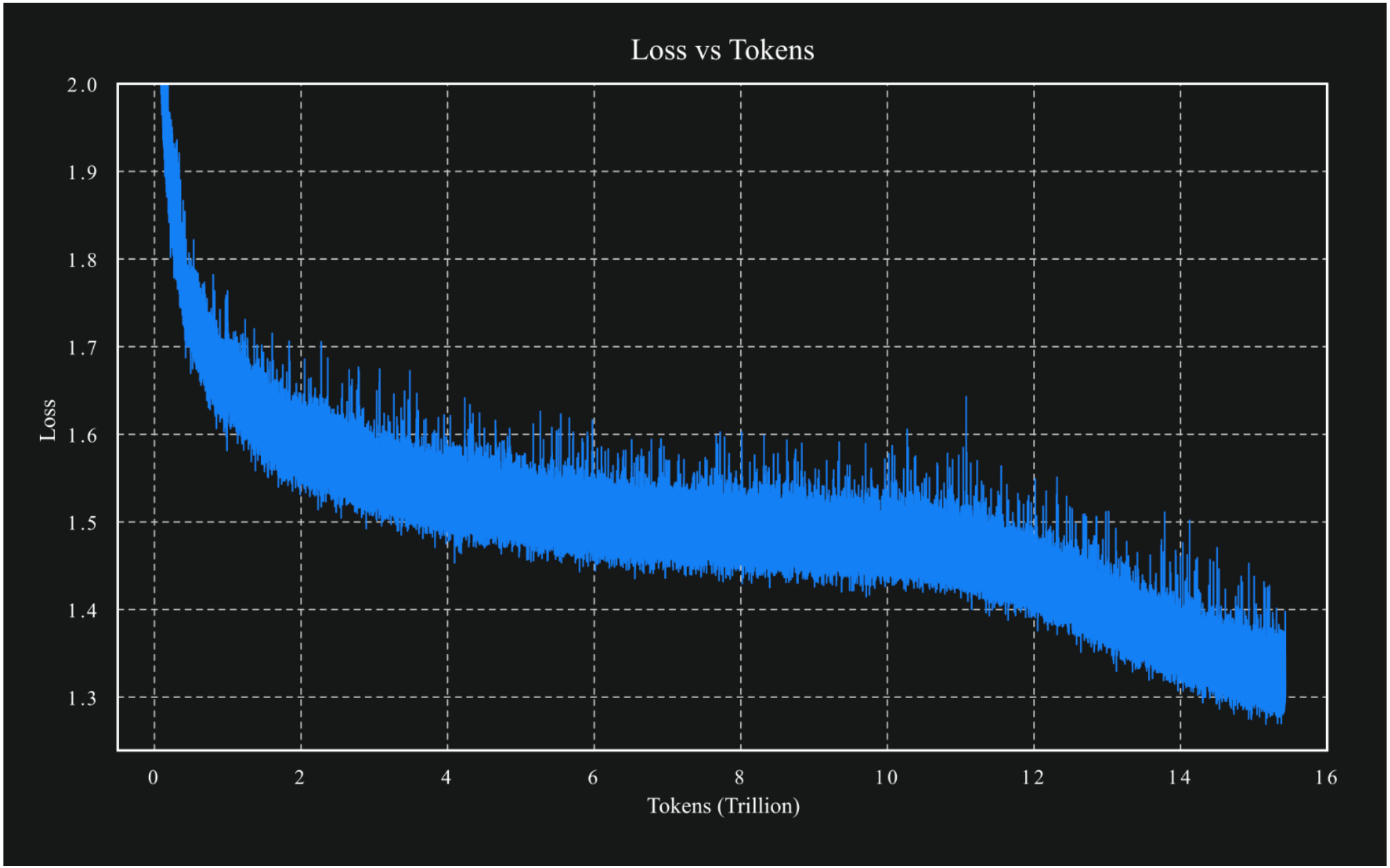
- vocabulary size (tokens): 160K
- embedding dimension: 7,168
- context length (tokens): 128K

- total parameters: 1T
- training tokens: 15.5T

→ a lot of memorizing

compression of the internet

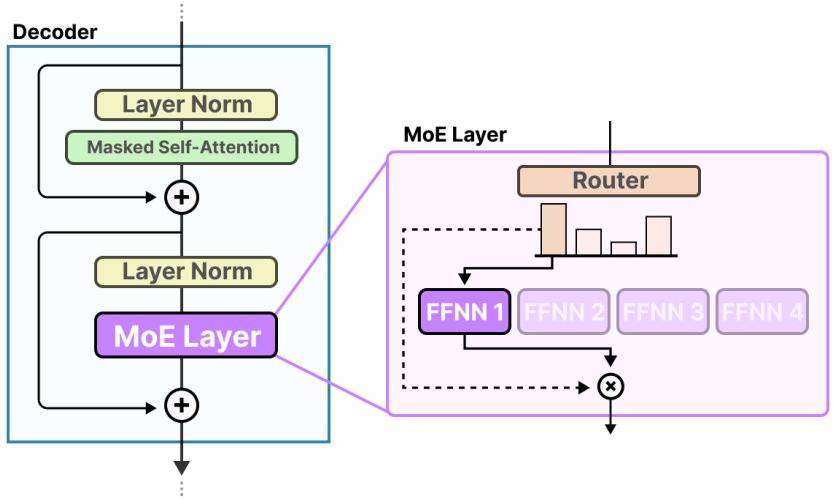
- number of layers: 61
- number of attention heads: 64
- number of experts: 384

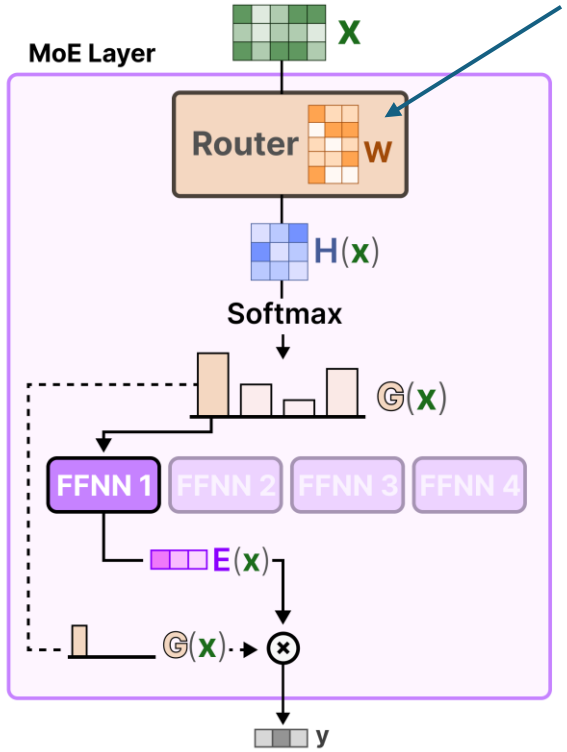

Loss vs Tokens
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
Loss
0
2
4
6
8
10
12
14
16
Tokens (Trillion)

source

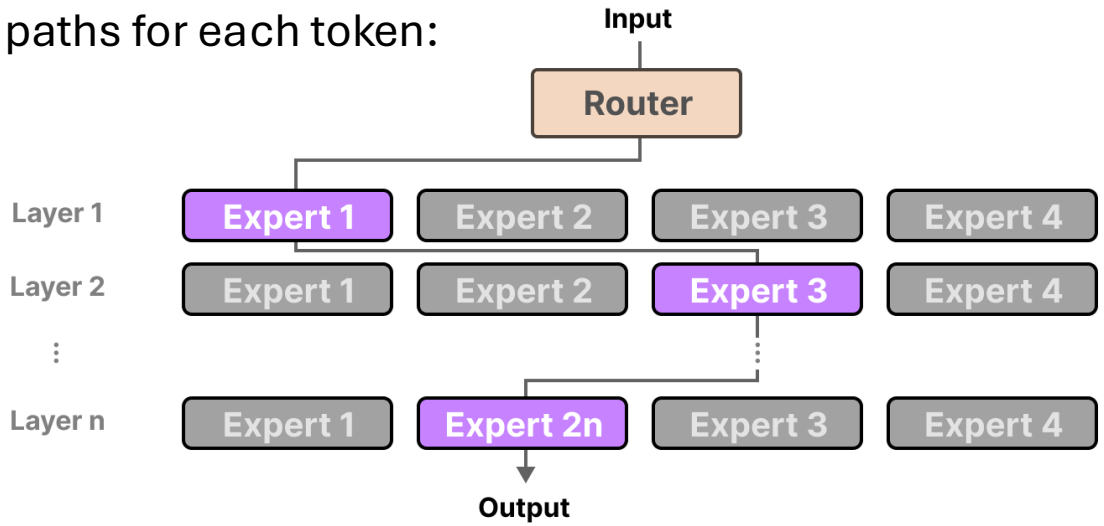
Mixture-of-Experts (MoE)

idea: replace big FFNN with several smaller ones and activate only few
→ less active parameters

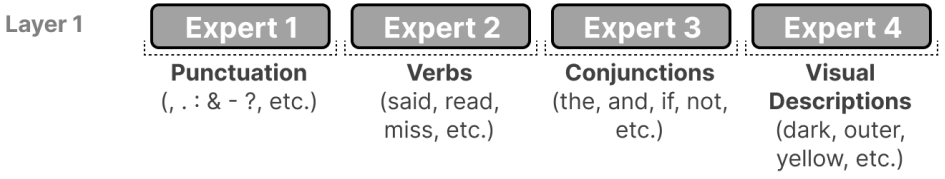


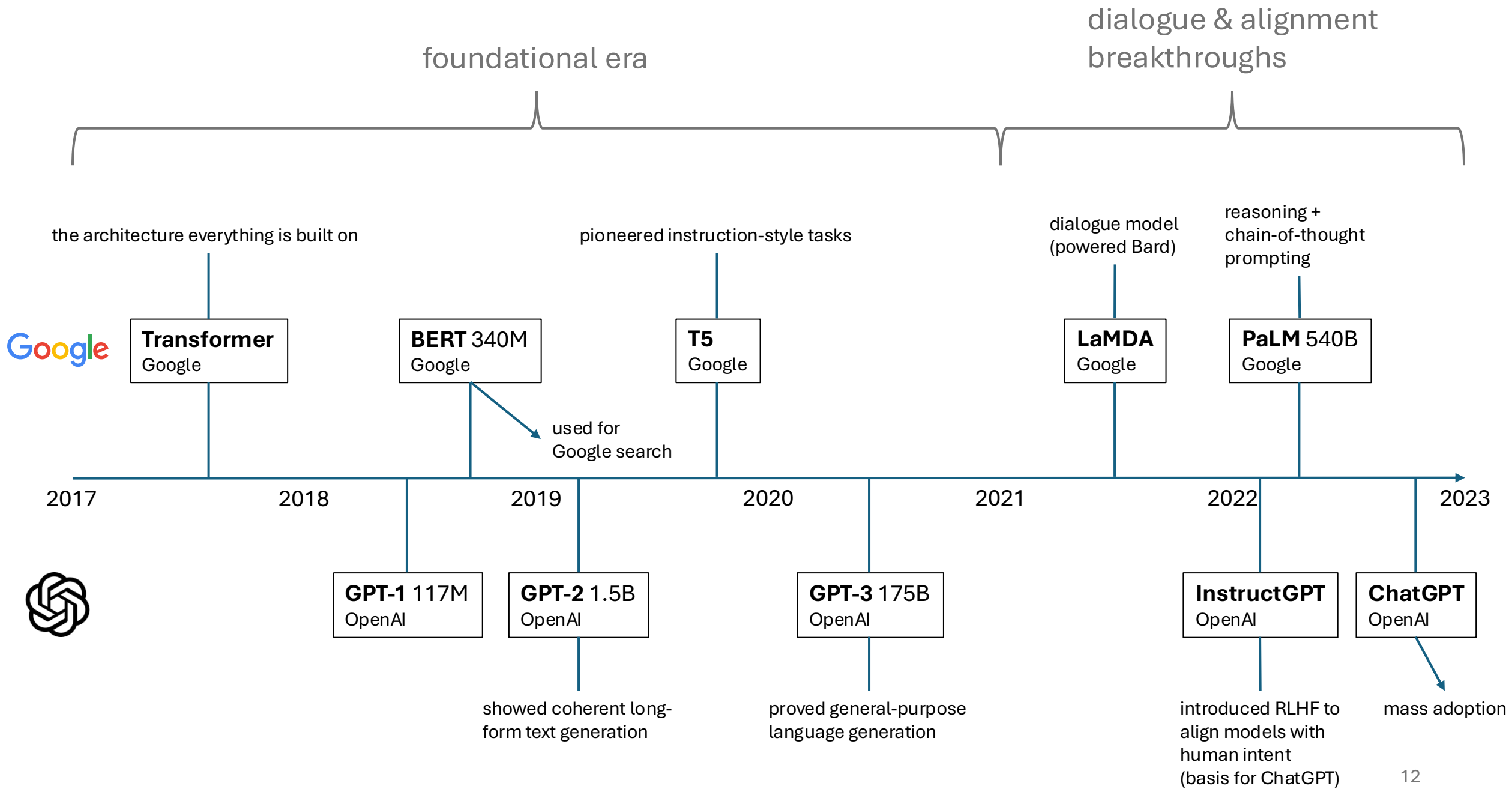


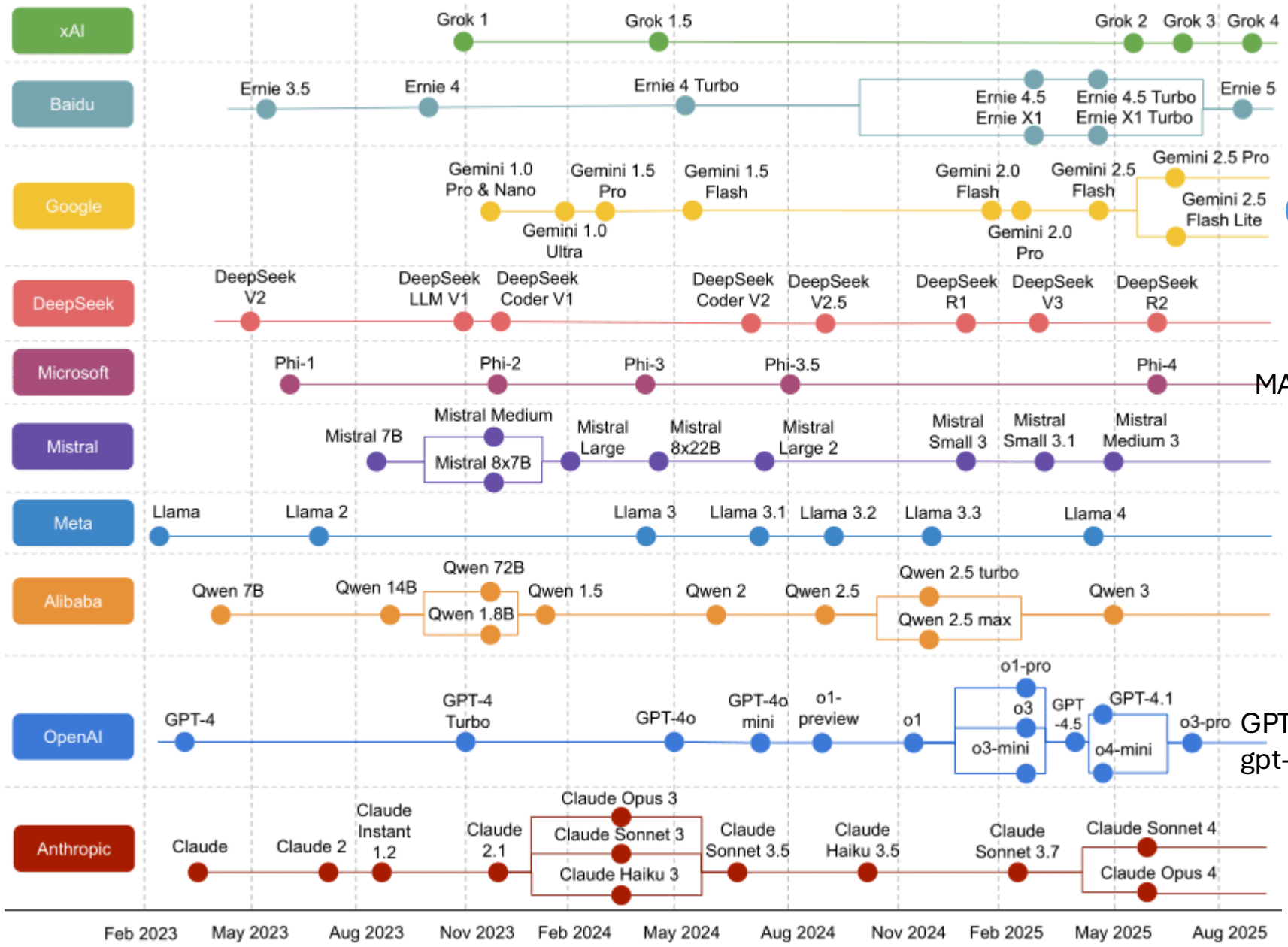
different paths for each token:



example (actually more abstract):




foundational era
dialogue & alignment breakthroughs
the architecture everything is built on
pioneered instruction-style tasks
dialogue model (powered Bard)
reasoning + chain-of-thought prompting
Google
Transformer Google
BERT 340M Google
used for Google search
T5 Google
LaMDA Google
PaLM 540B Google
2017
2018
2019
2020
2021
2022
2023
GPT-1 117M OpenAI
GPT-2 1.5B OpenAI
GPT-3 175B OpenAI
InstructGPT OpenAI
ChatGPT OpenAI
showed coherent long-form text generation
proved general-purpose language generation
introduced RLHF to align models with human intent (basis for ChatGPT)
mass adoption



generative chatbots

Gemini

- evolved from LaMDA & PaLM
- integrated Google ecosystem

LLaMA

open-source models

ChatGPT

Claude

- different alignment strategy
- tailored for business users

OpenAI's GPT Series

- GPT-1: used for discriminative finetuning on specific tasks (summarization, translation, question-answering, ...) → BERT
- GPT-2: zero- or few-shot learning → prompting
- GPT-3: scaling (175 billion parameters) → much increased capabilities
- GPT-4: more scaling (capabilities)
- GPT-4o: multimodal model
- o1: reasoning model
- GPT-5: automatic activation of reasoning model depending on the task

gpt-oss

LMArena

Rank (UB) ↑	Model	Score	95% CI (±)	Votes	Organization	License
1	gemini-2.5-pro	1455	±5	41.731	Google	Proprietary
1	claude-opus-4-1-20250805-thinking-16k	1451	±6	11.750	Anthropic	Proprietary
2	o3-2025-04-16	1444	±4	43.898	OpenAI	Proprietary
2	gpt-5-high	1442	±6	15.076	OpenAI	Proprietary
2	chatgpt-4o-latest-20250326	1441	±4	36.426	OpenAI	Proprietary
3	gpt-4.5-preview-2025-02-27	1439	±6	15.271	OpenAI	Proprietary
3	claude-opus-4-1-20250805	1438	±6	18.341	Anthropic	Proprietary
5	gpt-5-chat	1430	±6	11.808	OpenAI	Proprietary
6	qwen3-max-preview	1428	±7	8.781	Alibaba	Proprietary
8	grok-4-0709	1422	±5	21.446	xAI	Proprietary
8	claude-opus-4-20250514-thinking-16k	1420	±5	27.351	Anthropic	Proprietary
8	kimi-k2-0711-preview	1420	±5	23.527	Moonshot	Modified MIT
8	qwen3-235b-a22b-instruct-2507	1420	±6	17.947	Alibaba	Apache 2.0
8	deepseek-v3.1-thinking	1419	±8	8.226	DeepSeek	MIT
8	deepseek-v3.1	1418	±7	9.822	DeepSeek	MIT
8	deepseek-r1-0528	1418	±6	20.863	DeepSeek	MIT
8	kimi-k2-0905-preview	1418	±11	2.918	Moonshot	Modified MIT
10	glm-4.5	1411	±6	14.105	Z.ai	MIT

Another Trend: Small Language Models (SLM)

most powerful LLMs keep getting bigger (> 1 trillion parameters)
GPT, Claude, Gemini, …

but slimmed-down (yet still strong) versions get smaller
Gemma (2B), Phi-3 (3.8B), Mistral 7B, Llama3 (8B), …
can run (inference) on laptops, smartphones, etc.
easier finetuning → specific tasks

Next: Context Engineering

To get the most out of LLMs, you need to prompt in the right way.

→ prompt engineering

Furthermore, the capabilities of LLMs can be enhanced by including additional information in the prompt.

→ context engineering